DermAI

Skin Cancer Risk Classification System

GWU DATS 6303 · Deep Learning · Group 4 · Spring 2026

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10	13,799	0.67	76%
Classes	Images	Test F1	Accuracy

Problem & Motivation

1 in 5	Americans will develop skin cancer in their lifetime
5M+	New skin cancer cases diagnosed annually in the US
99%	5-year survival rate for melanoma when caught early
14-30	Days average wait time to see a dermatologist

Our Solution

- AI-powered skin lesion classification
- 10 skin condition categories
- Cancer Risk Score (0-100)
- Attention map for explainability
- Clinical mode for high-risk detection
- Instant results with report— no waiting

Dataset

HAM10000

- 10,015 dermoscopic images
- 7 lesion classes
- Harvard Dataverse

Kaggle Diseases

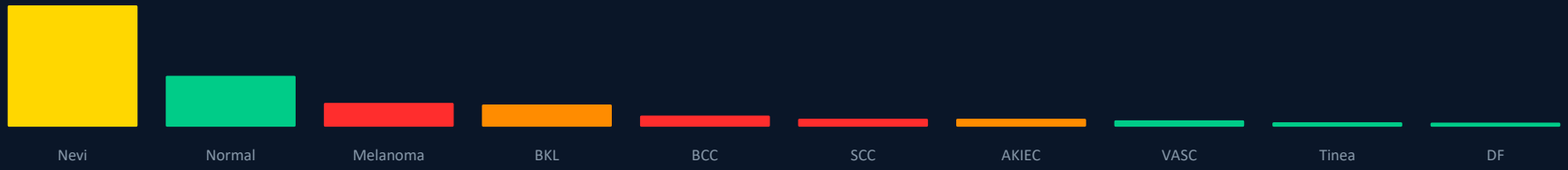
- SCC: 322 images
- Tinea: 122 images
- Healthy Skin: 2,756

ISIC 2020

- 584 melanoma images
- Train: 467 / Val: 117
- Address class imbalance

Final Dataset: 13,799 images · 10 classes · Train: 10,357 · Val: 2,096 · Test: 1,346

Class Distribution (Test Set)



Model Architecture

Input 224×224
RGB Image

Patch Split
16×16 patches
196 tokens

Linear Projection
768-dim embedding

Transformer Encoder ×12
12 attention heads

CLS Token
Classification token

MLP Head
10-class output

Why Vision Transformer?

- Global attention — captures lesion shape holistically
- Pre-trained on ImageNet-21k (14M images, 21k classes)
- Attention maps provide clinical explainability

86M	16×16	12	12	768
Parameters	Patch Size	Layers	Heads	Hidden

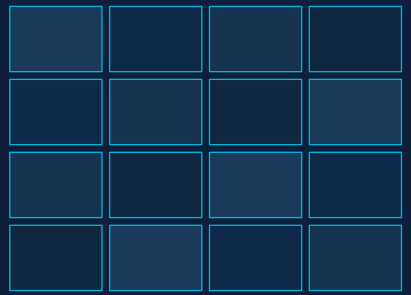
Baseline: EfficientNet-B3

- 42MB model — lightweight CNN baseline
- Compound scaling of depth, width, resolution
- Same training strategy and risk weights as ViT

12M	300×300	0.65	74%
Parameters	Input	F1	Accuracy

ViT Architecture — Deep Dive

① Patch Embedding



224×224 image
→ 196 patches of 16×16

Each patch → 768-dim vector
via linear projection

+ CLS token
+ Position embeddings

② Transformer Encoder ×12

Layer Norm

Multi-Head
Self-Attention

Layer Norm

MLP (FFN × 4)

12 heads × 64-dim = 768
Residual connections

③ Classification

CLS Token
768-dim

Linear Head
768 → 10

Softmax
Probabilities

Melanoma
BCC
SCC
...

EfficientNet-B3 — Baseline Architecture

Architecture Overview

Input 300×300×3

Stem Conv 3×3

MBConv1 ×2

MBConv6 ×3

MBConv6 ×3

MBConv6 ×4

MBConv6 ×4

MBConv6 ×5

Head + Classifier →10

Compound Scaling

EfficientNet scales depth, width, and resolution uniformly using compound coefficient ϕ

1.4×	1.2×	1.3×
Depth	Width	Resolution

MBConv Block

- 1×1 Conv (expand channels)
- 3×3 Depthwise Conv
- SE Block (squeeze & excite)
- 1×1 Conv (project)
- Residual connection

Training Strategy

Two-Phase Fine-tuning

Phase 1 — 5 epochs

Freeze encoder → train head only | LR: 1e-3

Phase 2 — 25 epochs

Unfreeze all → full fine-tuning | LR: 5e-5

Risk-Weighted Loss Function

Asymmetric cost: missing high-risk lesions is far more dangerous than false positives

8.0×	3.0×	2.5×	3.0×
Melanoma	BCC/AKIEC	SCC	DF

Other Training Details

- Label smoothing: 0.1
- CosineAnnealingLR scheduler
- WeightedRandomSampler
- CLAHE + ImageNet normalization
- Batch size: 32
- AdamW optimizer
- GPU: NVIDIA A10G (24GB)
- Training time: 0.5 hours

Results

Model Comparison

Model	Test F1	Accuracy	High Risk Recall
ViT-base (Standard)	0.6678	76%	68%
ViT-base (Clinical)	0.6338	70%	81%
EfficientNet (Standard)	0.6488	74%	58%
EfficientNet (Clinical)	0.6305	71%	76%

Key Achievements

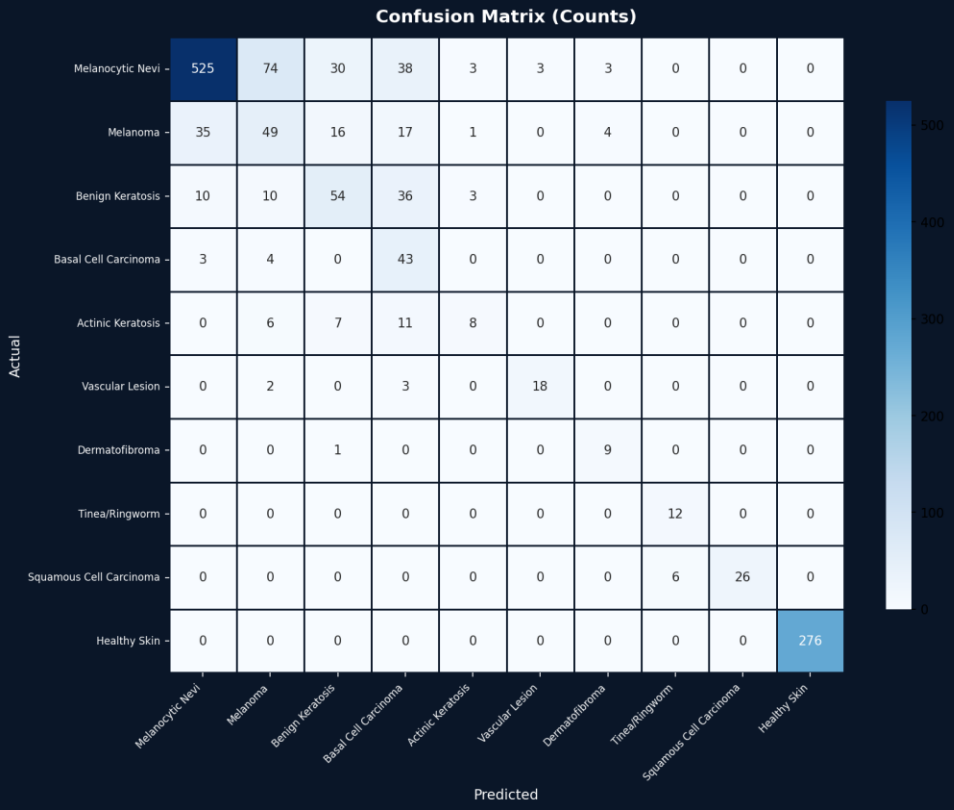
- ISIC2020 data: +467 melanoma training images
- ViT Clinical Mel Recall: 40% → 61%
- ENet Clinical Mel Recall: 43% → 62%

Clinical Mode Trade-off

Higher melanoma recall at cost of accuracy (-6%)

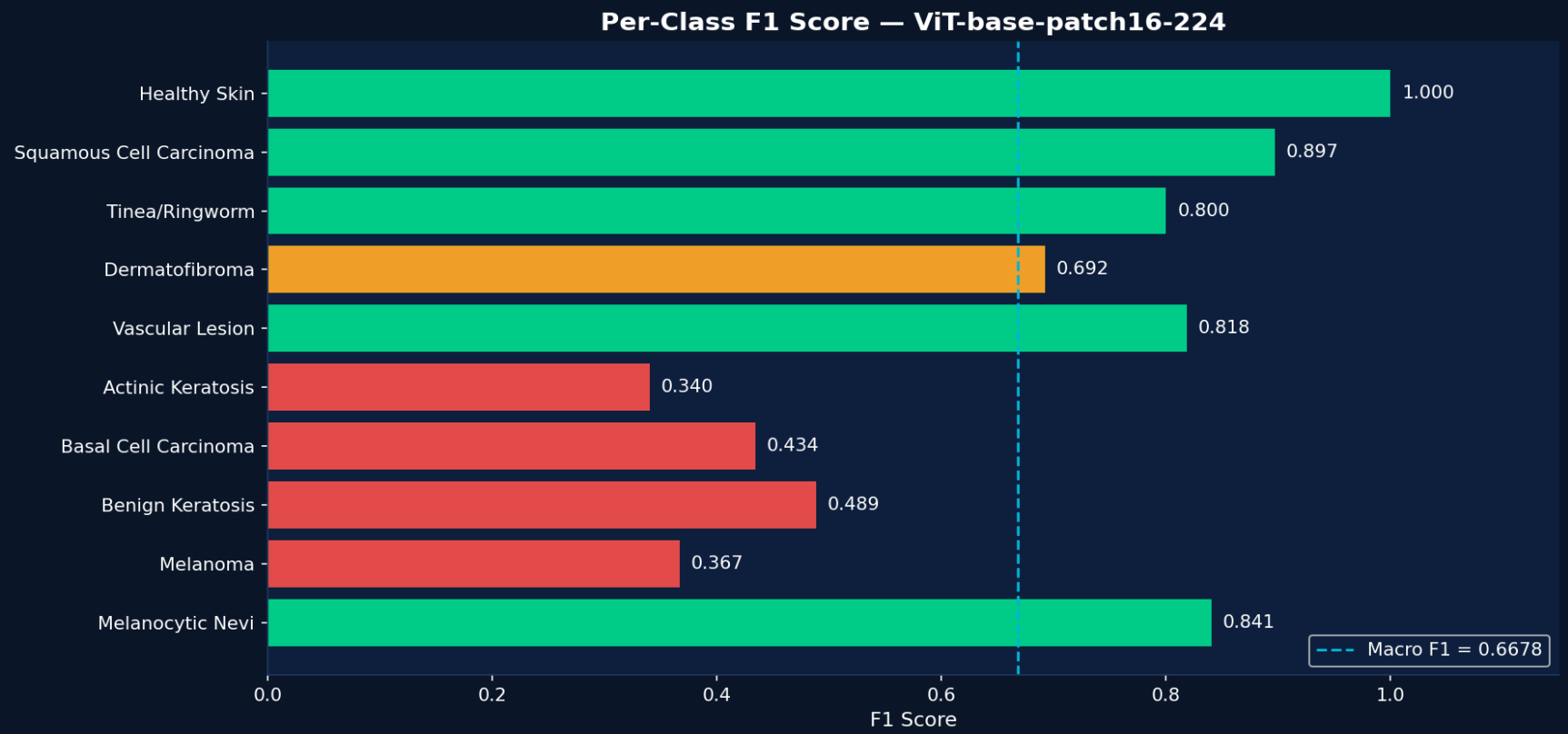
In clinical settings: missing a melanoma is far more dangerous than a false alarm

Confusion Matrix



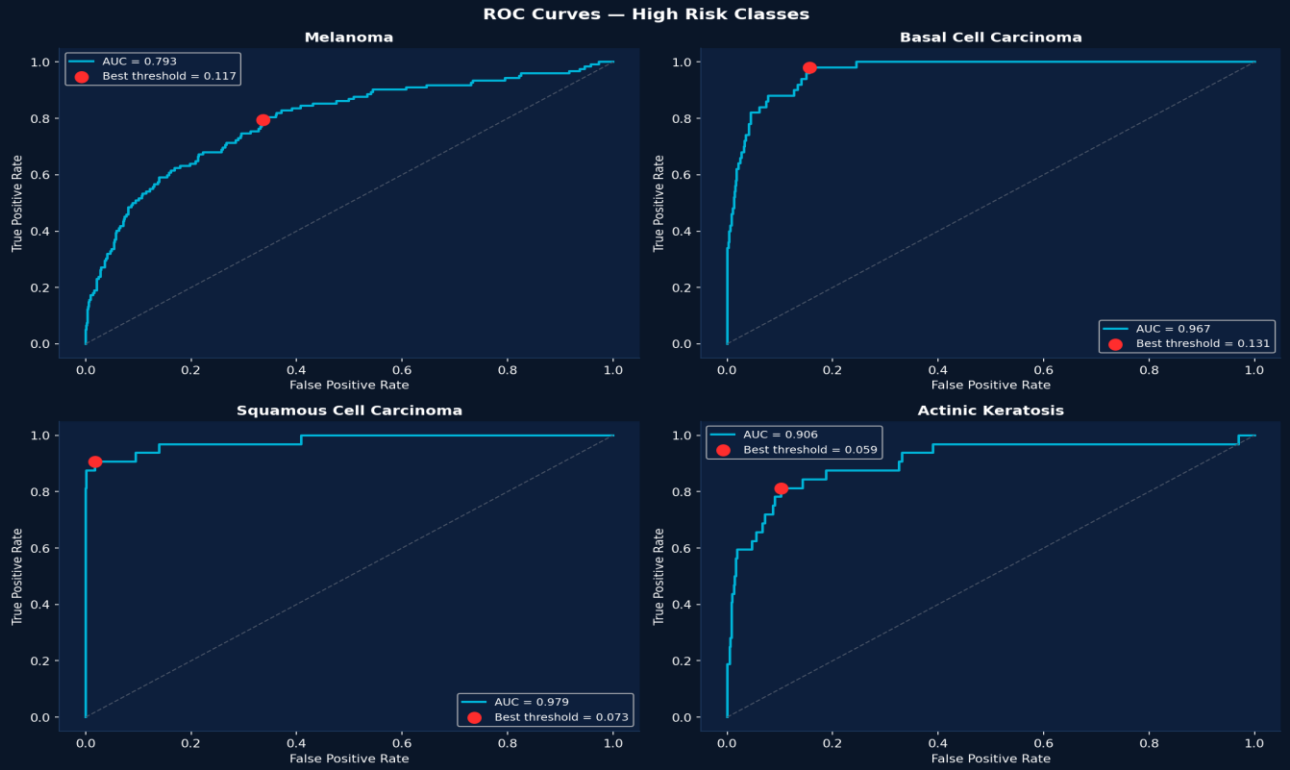
Rows = Actual class · Columns = Predicted class · Diagonal = Correct predictions

Per-Class F1 Score



Green = F1 ≥ 0.7 · Orange = F1 0.5-0.7 · Red = F1 < 0.5 · Dashed line = Macro F1

ROC Curves — Clinical Thresholds



Optimal Thresholds (Youden's J)

Melanoma	BCC	SCC	Actinic K.
t=0.117 TPR=80% AUC=0.793	t=0.131 TPR=98% AUC=0.967	t=0.074 TPR=91% AUC=0.979	t=0.059 TPR=81% AUC=0.906

Cancer Risk Score

$$\text{Risk Score} = \Sigma \left(P(\text{class_i}) \times \text{weight_i} \right) \times 100$$

Malignancy Weights

Melanoma	1.00
Squamous Cell Carcinoma	0.90
Basal Cell Carcinoma	0.80
Actinic Keratosis	0.65
Benign Keratosis	0.10
Melanocytic Nevi	0.10
Dermatofibroma	0.05
Vascular Lesion	0.05
Tinea/Ringworm	0.00
Healthy Skin	0.00

Clinical Mode & Explainability

Clinical Mode

Optimized thresholds via ROC analysis (Youden's J)
If P(class) ≥ threshold → boost to top prediction

	ViT Threshold	ENet Threshold
Melanoma	0.117	0.152
BCC	0.131	0.050
SCC	0.074	0.078
Actinic K.	0.059	0.121

Trade-off: +20% mel recall at cost of -6% accuracy

Attention Map Explainability

- ViT self-attention from last encoder layer
- Averaged across 12 attention heads
- Reshaped 14×14 → 224×224
- Overlaid as heatmap on original image
- Red/Yellow = high attention (lesion focus)
- Analogous to ABCDE criteria

AUC scores:

0.98	0.97	0.79
SCC	BCC	Mel

Live Demo

DermAI — Skin Cancer Risk Analyser

1

Select Demo Case

High / Elevated / Moderate / Low

2

Click SCAN

preprocessing + inference

3

View Risk Score

0-100 gauge + risk level

4

Toggle Clinical Mode

compare standard vs clinical

Conclusion & Future Work

What We Built

- 10-class skin condition classifier
- ViT-base fine-tuned on 13,215 images
- Cancer Risk Score with clinical weights
- Clinical mode with ROC-optimized thresholds
- Attention map explainability
- Full Streamlit demo application
- Mel Recall: 40% → 61% (clinical mode)

Future Work

- More melanoma data (ISIC 2019 full dataset)
- Ensemble ViT + EfficientNet predictions
- GradCAM for EfficientNet explainability
- Dermatologist validation study
- Mobile app deployment
- More diverse skin condition classes

10,357	10	0.668	76%	81%
Training Images	Classes	Test F1	Accuracy	High Risk Recall